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Table of Contents

1 Introduction 1
1.1 Aim 1
1.2 Objective 1
1.3 Required tools and concepts 1
1.4 Steps Required for Labs 2
1.4.1 Step-1 2
1.4.2 Step-2 3
1.4.3 Step-3 3
1.4.4 Step-4 4
1.4.5 Step-5 4
1.4.6 Step-6 5
1.4.7 Step-7 5
1.4.8 Step-8 6
1.4.9 Step-9 7
1.4.10 Step-10 8
1.4.11 Step-11 9
1.5 Conclusion 11

Table of figures

Figure 1 2
Figure 2 2
Figure 3 3
Figure 4 3
Figure 5 4
Figure 6 4
Figure 7 5
Figure 8 6
Figure 9 6
Figure 10 7
Figure 11 8
Figure 12 10

1 Introduction

1.1 Aim

The main aim of this lab is to enhance and upgrade the skill in Linux basic commands like managing directories, files and permission in Debian.

1.2 Objective

- To create a folder structure using a Linux basic command.
- Changing directories in directory structure by one command using relative pathname.
- To copy the text file from one folder to another folder with the same as in previous folder.
- To print the text in text file using echo and printf command.
- To read the file using an utility cat.

1.3 Required tools and concepts

Tools:

- Linux Operating System (Debian)
- Command line Interface.

Concepts:

- Windows Subsystem for Linux

1.4 Steps Required for Labs

1.4.1 Step-1

Create the directory structure presented in the figure below. Use mkdir command and relative pathnames from your home directory. Try both no option and -p option, for the command.

```
manoj@DESKTOP-I4F9RFA:~$ mkdir -p w7/{w7-1/{1level3,2level3},w7-2/{3level3,4level3}}
manoj@DESKTOP-I4F9RFA:~$ tree
.
├── alscript
└── w7
    ├── w7-1
    │   ├── 1level3
    │   └── 2level3
    └── w7-2
        ├── 3level3
        └── 4level3

8 directories, 1 file
manoj@DESKTOP-I4F9RFA:~$
```

Figure 1

```
manoj@DESKTOP-I4F9RFA:~$ mkdir w7
manoj@DESKTOP-I4F9RFA:~$ mkdir w7/w7-1
manoj@DESKTOP-I4F9RFA:~$ mkdir w7/w7-2
manoj@DESKTOP-I4F9RFA:~$ mkdir w7/w7-1/1level3
manoj@DESKTOP-I4F9RFA:~$ mkdir w7/w7-1/2level3
manoj@DESKTOP-I4F9RFA:~$ mkdir w7/w7-2/3level3
manoj@DESKTOP-I4F9RFA:~$ mkdir w7/w7-2/4level3
manoj@DESKTOP-I4F9RFA:~$ tree w7
w7
├── w7-1
│   ├── 1level3
│   └── 2level3
└── w7-2
    ├── 3level3
    └── 4level3

7 directories, 0 files
manoj@DESKTOP-I4F9RFA:~$
```

Figure 2

1.4.2 Step-2

Change to the 1level3 directory by one step using a relative pathname.

```
manoj@DESKTOP-I4F9RFA:~$ cd w7/w7-1/1level3
manoj@DESKTOP-I4F9RFA:~/w7/w7-1/1level3$
```

Figure 3

1.4.3 Step-3

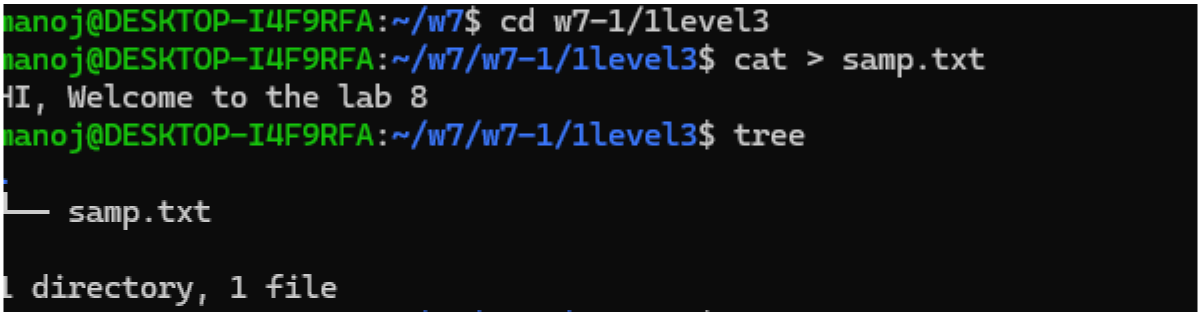
Practice in changing directories in your directory structure by one command and using relative pathnames e.g. from 1level3 to 2 level3, from 2level3 to 4 level3, from 4level3 to w7 etc. Use names of parent and child directories as well.

```
manoj@DESKTOP-I4F9RFA:~$ cd w7/w7-1/1level3
manoj@DESKTOP-I4F9RFA:~/w7/w7-1/1level3$ cd ../2level3
manoj@DESKTOP-I4F9RFA:~/w7/w7-1/2level3$ cd ../../w7-2/4level3
manoj@DESKTOP-I4F9RFA:~/w7/w7-2/4level3$ cd ../../
manoj@DESKTOP-I4F9RFA:~/w7$
```

Figure 4

1.4.4 Step-4

Change to 1level3 and create a text file by any tool (e.g. by cat or cal)

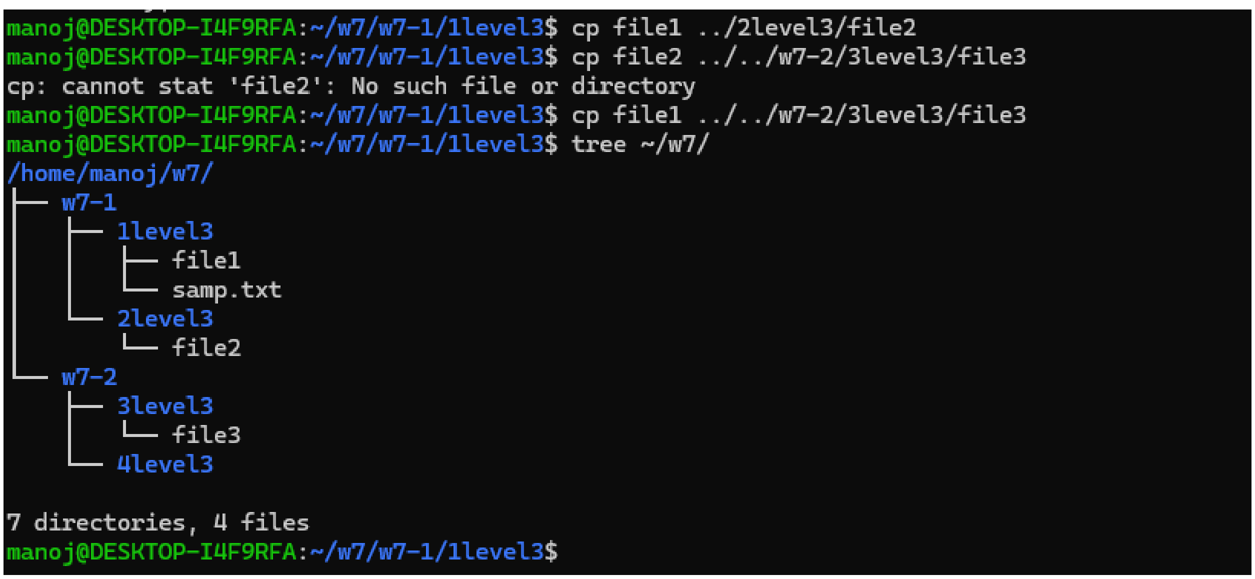


```
anoj@DESKTOP-I4F9RFA:~/w7$ cd w7-1/1level3
anoj@DESKTOP-I4F9RFA:~/w7/w7-1/1level3$ cat > samp.txt
HI, Welcome to the lab 8
anoj@DESKTOP-I4F9RFA:~/w7/w7-1/1level3$ tree

└── samp.txt

 directory, 1 file
```

Figure 5

1.4.5 Step-5

Copy this text file from 1level3 to 1level3 with the name file1, 2level3 and to 3level3 changing its name. Show that there are these files in corresponding directories.



```
manoj@DESKTOP-I4F9RFA:~/w7/w7-1/1level3$ cp file1 ../2level3/file2
manoj@DESKTOP-I4F9RFA:~/w7/w7-1/1level3$ cp file2 ../../w7-2/3level3/file3
cp: cannot stat 'file2': No such file or directory
manoj@DESKTOP-I4F9RFA:~/w7/w7-1/1level3$ cp file1 ../../w7-2/3level3/file3
manoj@DESKTOP-I4F9RFA:~/w7/w7-1/1level3$ tree ~/w7/
/home/manoj/w7/
├── w7-1
│   ├── 1level3
│   │   ├── file1
│   │   └── samp.txt
│   └── 2level3
│       └── file2
└── w7-2
    ├── 3level3
    │   └── file3
    └── 4level3

7 directories, 4 files
manoj@DESKTOP-I4F9RFA:~/w7/w7-1/1level3$
```

Figure 6

1.4.6 Step-6

Move this file to 4level3. Show that there is this file in 4level3 and there is not in 1level3.

```
manoj@DESKTOP-I4F9RFA:~/w7/w7-1/1level3$ tree ~/w7/
/home/manoj/w7/
├── w7-1
│   ├── 1level3
│   │   ├── file1
│   │   └── samp.txt
│   └── 2level3
│       └── file2
└── w7-2
    ├── 3level3
    │   └── file3
    └── 4level3

7 directories, 4 files
manoj@DESKTOP-I4F9RFA:~/w7/w7-1/1level3$ mv file1 ../../w7-2/4level3/file
manoj@DESKTOP-I4F9RFA:~/w7/w7-1/1level3$ tree
.
└── samp.txt

1 directory, 1 file
manoj@DESKTOP-I4F9RFA:~/w7/w7-1/1level3$ tree ~/w7/
/home/manoj/w7/
├── w7-1
│   ├── 1level3
│   │   └── samp.txt
│   └── 2level3
│       └── file2
└── w7-2
    ├── 3level3
    │   └── file3
    └── 4level3
        └── file

7 directories, 4 files
manoj@DESKTOP-I4F9RFA:~/w7/w7-1/1level3$
```

Figure 7

1.4.7 Step-7

Print the following texts each in one echo or printf command.

- Hello! I can do it
- 5 > (20:8) < (30 *2)
- Line 1

  Line 2

  a-b,A-B,-,+,<,>,#,$,%,&.

```
manoj@DESKTOP-I4F9RFA:~/w7/w7-1/1level3$ echo -e "Hello! I can do it\n5 > (20:8) < (30 * 2)\nLine1\nLine2\na-b,A-B,-,+,<,>,#,$,%,&"
Hello! I can do it
5 > (20:8) < (30 * 2)
Line1
Line2
a-b,A-B,-,+,<,>,#,$,%,&
manoj@DESKTOP-I4F9RFA:~/w7/w7-1/1level3$
```

Figure 8

1.4.8 Step-8

Give the command without options and with a,d,g,I,R options in home directory. w7,w7-1, and 1level3 directories.

```
manoj@DESKTOP-I4F9RFA:~/w7/w7-1/1level3$ cd ~
manoj@DESKTOP-I4F9RFA:~$ ls
alscript  w7
manoj@DESKTOP-I4F9RFA:~$ ls -a
.  ..  alscript  .bash_history  .bash_logout  .bashrc  .profile  .sudo_as_admin_successful  w7
manoj@DESKTOP-I4F9RFA:~$ ls -d w7
w7
manoj@DESKTOP-I4F9RFA:~$ ls -d ls -d w7/w7-1/1level3
ls: cannot access 'ls': No such file or directory
w7/w7-1/1level3
manoj@DESKTOP-I4F9RFA:~$ ls -d w7/w7-1/1level3
w7/w7-1/1level3
manoj@DESKTOP-I4F9RFA:~$ ls -g
total 12
-rw-r--r-- 1 manoj 7744 Dec  7 03:27 alscript
drwxr-xr-x 4 manoj 4096 Dec 16 02:47 w7
manoj@DESKTOP-I4F9RFA:~$ ls -I "*.txt"
alscript  w7
manoj@DESKTOP-I4F9RFA:~$ ls -R w7
w7:
w7-1  w7-2

w7/w7-1:
1level3  2level3

w7/w7-1/1level3:
samp.txt

w7/w7-1/2level3:
file2
```

Figure 9

1.4.9 Step-9

Change to the w7 directory. Remove the directory files w7-2, 3level3,4level3 and all ordinary files in them. Use the option -I of the rm and rmdir commands. Show that there are not these ordinary and directory files in your file structure.

```
manoj@DESKTOP-I4F9RFA:~$ cd ~/w7
manoj@DESKTOP-I4F9RFA:~/w7$ cd w7-2
manoj@DESKTOP-I4F9RFA:~/w7/w7-2$ rm -i 3level3/*
rm -i 4level3/*
rm: remove regular file '3level3/file3'? y
rm: remove regular file '4level3/file'? y
manoj@DESKTOP-I4F9RFA:~/w7/w7-2$ rmdir -i 3level3
rmdir -i 4level3
rmdir: invalid option -- 'i'
Try 'rmdir --help' for more information.
rmdir: invalid option -- 'i'
Try 'rmdir --help' for more information.
manoj@DESKTOP-I4F9RFA:~/w7/w7-2$ rmdir 3level3
rmdir 4level3
manoj@DESKTOP-I4F9RFA:~/w7/w7-2$ rm -ri 3level3
rm -ri 4level3
rm: cannot remove '3level3': No such file or directory
rm: cannot remove '4level3': No such file or directory
manoj@DESKTOP-I4F9RFA:~/w7/w7-2$ tree
.

0 directories, 0 files
manoj@DESKTOP-I4F9RFA:~/w7/w7-2$ tree ~/w7
/home/manoj/w7
├── w7-1
│   ├── 1level3
│   │   └── samp.txt
│   └── 2level3
│       └── file2
└── w7-2

5 directories, 2 files
manoj@DESKTOP-I4F9RFA:~/w7/w7-2$ cd ~/w7
rmdir w7-2
manoj@DESKTOP-I4F9RFA:~/w7$
```

Figure 10

1.4.10 Step-10

Change to **w7-1**.

- Display access permissions for file **file1** in **1level3**.
- Remove all access permissions for this file.
- Display access permissions for this file.
- Try to read this file using any utility (e.g.,**cat**).
- Try to write into this file using any utility (e.g., **cat** with the sign **>>** – *append*).
- Add read and write access permissions for yourself for this file.
- Display access permissions for this file.
- Try to read this file using any utility.
- Try to write into this file using any utility.

```
manoj@DESKTOP-I4F9RFA:~/w7/w7-2$ tree ~/w7
/home/manoj/w7
├── w7-1
│   ├── 1level3
│   │   └── samp.txt
│   └── 2level3
│       └── file2
└── w7-2

5 directories, 2 files
manoj@DESKTOP-I4F9RFA:~/w7/w7-2$ cd ~/w7
rmdir w7-2
manoj@DESKTOP-I4F9RFA:~/w7$ cd ..
manoj@DESKTOP-I4F9RFA:~$ cd ~/w7/w7-1
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ ls -l 1level3/file1.txt
ls: cannot access '1level3/file1.txt': No such file or directory
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ w7/w7-1/1level3/samp.txt
-bash: w7/w7-1/1level3/samp.txt: No such file or directory
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ w7/w7-1/1level3/samp.txt
-bash: w7/w7-1/1level3/samp.txt: No such file or directory
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ ls -l 1level3/samp.txt
-rw-r--r-- 1 manoj manoj 26 Dec 16 03:04 1level3/samp.txt
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ chmod 000 1level3/samp.txt
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ ls -l 1level3/samp.txt
---------- 1 manoj manoj 26 Dec 16 03:04 1level3/samp.txt
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ cat >> 1level3/samp.txt
-bash: 1level3/samp.txt: Permission denied
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ chmod u+rw 1level3/samp.txt
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ ls -l 1level3/samp.txt
-rw------- 1 manoj manoj 26 Dec 16 03:04 1level3/samp.txt
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ cat 1level3/samp.txt
HI, Welcome to the lab 8
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ cat >> 1level3/samp.txt
hello guys welcome to debian
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ |
```

Figure 11

1.4.11 Step-11

- Display access permissions for **1level3**.
- Remove all access permissions for the
- **1level3** directory.
- Display access permissions for **1level3**.
- Try to read a file from **1level3** using any utility.
- Try to put a file into **1level3** using any utility.
- Try to search in **1level3** using any command (e.g., the **ls** command).
- Add read, write, and execute access permissions for yourself for the **1level3** directory.
- Display access permissions for **1level3**.
- Try to read a file from **1level3** using any utility.
- Try to put a file into **1level3** using any utility.
- Try to search in **1level3** using any command (e.g., the **ls** command).

```
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ tree ~/w7
/home/manoj/w7
└── w7-1
    ├── 1level3
    │   └── samp.txt
    └── 2level3
        └── file2

4 directories, 2 files
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ ls -ld 1level3
drwxr-xr-x 2 manoj manoj 4096 Dec 16 03:30 1level3
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ chmod 000 1level3
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ ls -ld 1level3
d--------- 2 manoj manoj 4096 Dec 16 03:30 1level3
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ cat 1level3/samp.txt
cat: 1level3/samp.txt: Permission denied
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ cat > 1level3/test.txt
-bash: 1level3/test.txt: Permission denied
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ ls 1level3
ls: cannot open directory '1level3': Permission denied
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ chmod u+rwx 1level3
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ ls -ld 1level3
drwx------ 2 manoj manoj 4096 Dec 16 03:30 1level3
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ cat 1level3/samp.txt
HI, Welcome to the lab 8
hello guys welcome to debian
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ cat > 1level3/test.txt
Heelo network operating system
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$ ls 1level3
samp.txt  test.txt
manoj@DESKTOP-I4F9RFA:~/w7/w7-1$
```

Figure 12

1.5 Conclusion

In this lab, Linux file and directory management command were practiced using relative and absolute pathnames. Directory is created using mkdir, ls and tree. Files were created, copied and removed using cat, cp, rm. Rmidir. Files access permissions were practice using chmod command.